

CHAPTER 6

Antenna Terminology

6.1. INTRODUCTION

Antennas are basic components of any electric system and are connecting links between the transmitter and free-space or free space and the receiver. Thus antennas play very important role in finding the characteristics of the system in which antennas are employed. Antennas are employed in different systems in different forms. That is, in some systems the operational characteristic of the system are designed around the directional properties of the antennas or in some other systems, the antennas are used simply to radiate electromagnetic energy in an omnidirection or finally in some systems for point to point communication purposes in which increased gain and reduced wave interference are required.

Irrespective of type of application of an antennas or antenna system, all the antennas possessed certain ^{basic} properties which will now be discussed in this chapter. The important properties of an antenna are ^{generally} given in some terms like, Radiation Pattern, Radiation Intensity, Polarization gain, directive ^{gain} or directivity, Power gain, Efficiency, Effective Aperture or Area, Self and Mutual impedance, radiation resistance, Beam width, Band width etc. These antenna terminology will one by one be taken up. ^{however} however, we shall start with another term Isotropic Radiator which will be needed subsequently for comparison purposes.

6.2. ISOTROPIC RADIATORS

(AMIETE, Nov. 1969)

An isotropic radiator is a fictitious radiator and is defined as a radiator which radiates uniformly in all directions. It is also called as isotropic source or omnidirectional radiator or simply unipole. An isotropic radiator is a hypothetical lossless radiator or antenna, with which the practical radiators or antennas are compared. Thus an isotropic antenna or radiator is used as reference antenna. Although, sometimes, a half-wave dipole antenna is also used as reference antenna but these days use of isotropic antenna as reference antenna is preferred.

Since all the practical antennas have atleast some directional properties i.e. directivity and hence there is no such thing as isotropic radiator of electromagnetic energy. However, in acoustics we have and a point source of sound is an example of isotropic radiator.

Let us now imagine that an isotropic radiator is situated at the centre of a sphere of radius (r). Then all the energy (power) radiated from it, must pass over the surface area of the sphere (assuming there is no

(obstruction to absorb the power) $4 \pi r^2$. Further, as said earlier, the Poynting vector or power density $\mathbf{P}$ at any point on the sphere gives "Power radiated per unit area in any direction". Since radiated power from a isotropic source flows in radial lines, therefore, for a isotropic radiator, the magnitude of the Poynting vector $\mathbf{P}$ is equal to the radial component only (because $P_\theta = P_\phi = 0$).

$$\therefore |\mathbf{P}| = P_r \quad \dots (6.1)$$

Thus, if the Poynting vector is known at all points on a sphere of radius r from a point source in a lossless medium, the total power (W_t) radiated by the source is integral over the surface of the sphere of the radial component P_r of average Poynting vector. Symbolically,

$$\begin{aligned} W_t &= \iint \mathbf{P} \cdot d\mathbf{s} \\ &= \iint P_r \cdot d\mathbf{s} = P_r \iint d\mathbf{s} \quad \text{from 6.1 } \because \theta = \phi = 0 \text{ for isotropic radiator} \end{aligned} \quad \dots (6.2)$$

$$W_t = P_r 4 \pi r^2 \quad \therefore \int \mathbf{P} = P_r \because \int d\mathbf{s} = 4 \pi r^2 = \text{area of surface} \quad \dots 6.3 (a)$$

or

$$P_r = \frac{W_t}{4 \pi r^2} \text{ Watts/m}^2$$

... 6.3 (b)

Here

W_t = Total power radiated, in Watts.

P_r = Radial component of average power density poynting vector, in W/m^2 .

$d\mathbf{s}$ = Infinitesimal element of area of sphere of radius $r = r^2 \sin \theta d\theta d\phi$.

and

r = Radius of sphere, in metres.

6.3. RADIATION PATTERN (V/m) or (mV/m) (AMIETE, May 1977)

It is found, in practice, that radiated energy from an antenna is not of the same strength in all directions. Instead, it is more in one direction and less or zero in other direction. The energy radiated in a particular direction by an antenna is measured in terms of FIELD STRENGTH at a point which is at a particular distance from the antenna.

For the calculation of field strength, the voltages at two points on an electric line of force, are taken and then it is divided by the distance between the two points. Hence the unit of radiation pattern is volt/metre or milli-volt/metre. The radiation pattern of an antenna is generally its most basic requirement because it determines the distribution of radiated energy in space. Once the operating frequency is known, the radiation pattern is the first property of an antenna that is specified.

Radiation Pattern of an antenna is nothing but a graph which shows the variation in actual field strength of electromagnetic field at all points which are at equal distance from the antenna. Obviously the graph of radiation pattern will be three-dimensional and hence can not completely be represented on a plain paper. In order to draw the radiation pattern of an antenna field strengths are measured at every point which lies on the surface of an imaginary sphere of a fixed radius treating antenna as centre and then a three dimensional solid figure is constructed from the reading so obtained. In this case distance from the centre (fixed point) to the surface of sphere represents the field strength in that direction. In order that the radiation pattern of an antenna is represented on a plain paper, the three dimensional solid figure so obtained is cut by a plane passing through the fixed point (say centre of sphere) and the figures now obtained are used to represent the radiation pattern in a usual way. The radiation patterns are different for different antennas and are affected by the location of antenna w.r.t. ground.

In fact, the graphical representation of radiation of an antenna as a function of direction is given the name radiation pattern of the antenna. If the radiation from the antenna is expressed in terms of field strength E (volt/metre), the radiation pattern is called as the 'Field Strength Pattern'. If, on other hand, the radiation in a given direction is expressed in terms of power per unit solid angle, then the resulting pattern is a 'Power Pattern'. However, both are related to each other—a power pattern is proportional to the

square of the field strength pattern — and unless otherwise mentioned radiation patterns henceforth referred to will be field strength pattern.

Since the radiation pattern is a three dimensional figure and hence the co-ordinate system usually used for the same is the 'spherical co-ordinate' (r, θ, ϕ) . The antenna is assumed to be located at origin of spherical co-ordinate system and the field strength is specified at points on the spherical surface of radius (r) . The shape of the radiation pattern does not depend on the radius r provided $r \gg \lambda$.

The direction of field strength (E) for the radiation field is always tangential to the spherical surface of imaginary sphere of radius r and for vertical dipole electric field strength E is in the direction θ and for the horizontal loop in the direction of ϕ .

In other words, radiation field strength may have components E_θ and E_ϕ as well which may or may not be in time phase or mathematically,

$$E = \sqrt{E_\theta^2 + E_\phi^2} \quad \dots (6.4)$$

where

E = Total electric field strength.

E_θ = Amplitude of θ component.

E_ϕ = Amplitude of ϕ component.

The radiation characteristics for vertically and horizontal polarizations are different and hence shown by different radiation patterns.

A complete radiation pattern is a three dimensional solid figure and gives the radiation for all angles of the θ and ϕ . However to represent the radiation pattern on a plain paper (i.e. in two dimension) a cross-section through three dimensional pattern is taken. Cross-sections generally taken are in horizontal plane (when $\theta = 90^\circ$) and in vertical plane (when $\phi = \text{constant}$). Thus the two dimensional patterns so obtained from three-dimensional pattern by cutting with a horizontal and vertical planes are, respectively, known as 'Horizontal Pattern' and 'Vertical Pattern'.

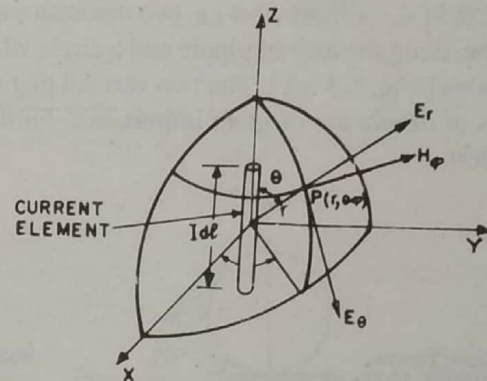


Fig. 6.1. Elementary dipole at the centre spherical co-ordinate system

Let us now come to see the radiation patterns in particular. For example, an elementary dipole antenna (Idl). The magnitude of 'radiation term' for such an antenna is given by

$$E_\theta = \frac{60 \pi Idl}{r \lambda} \cdot \sin \theta \quad \dots 6.5 (a)$$

or

$$E_\theta \propto \sin \theta \quad \dots 6.5 (b)$$

where

θ = Angle between the axis of the dipole and the radius vector to the point where the field strength E is measured (Fig. 6.2).

Thus as seen from eqn. 6.5 (b) E_θ (and so also the H_ϕ) is proportional to $\sin \theta$. Therefore, the field strength is maximum when $\theta = 90^\circ$ (because $\sin 90^\circ = 1$) in the $\perp$ direction of axis of dipole and minimum when $\theta = 0$ (because $\sin 0^\circ = 0$) in the direction of axis of dipole. The variation of E_θ (or H_ϕ) with angle can be drawn by a field pattern as shown in Fig. 6.2 (b).

Fig. 6.2 (a) is one half of a three dimensional pattern illustrates that fields are functions of θ and does not depend on the angle while Fig. 6.2 (b) is a two dimensional pattern obtained by cutting the three-dimensional pattern by a vertical plane along the axis of dipole. In other words Fig. 6.2 (b) is the front view of Fig. 6.2 (a). Fig. 6.2 (c) is also two dimensional pattern obtained by cutting three dimensional pattern by a horizontal plane at the centre of the dipole. As seen, three dimensional pattern of a short dipole is Doughnut

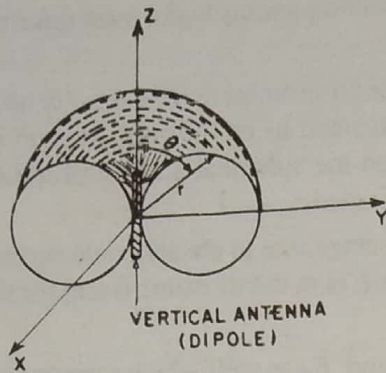


Fig. 6.2. (a) Half of the three-dimensional pattern (Doughnut shape)

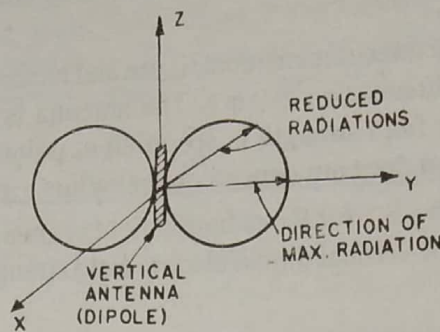


Fig. 6.2 (b) Two dimensional pattern obtained by cutting three dimensional pattern with a vertical plane along the axis of dipole (vertical pattern)

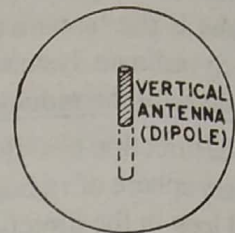


Fig. 6.2 (c) Two dimensional pattern when cut by a horizontal plane at the centre of dipole

shaped [Fig. 6.2 (a)], two dimensional pattern is a figure of eight ∞ [Fig. 6.2 (b)] when cut by a vertical plane and a circle [Fig. 6.2 (c)] when cut by a horizontal plane. Above patterns are of course, for a vertically placed dipole antenna. Similarly, for a horizontally placed dipole antenna, three dimensional pattern is as shown in Fig. 6.3 (a), its front view i.e. two dimensional pattern is a figure of eight 8 [Fig. 6.3 (b)] when cut by a vertical plane along the axis of dipole and a circle when cut by a vertical plane at the centre of dipole perpendicular to its axis [Fig. 6.3 (c)]. The two vertical planes i.e. one along the axis of dipole and the other perpendicular to axis of dipole are of great importance. Further increase in length of antenna changes the shape of radiation pattern.

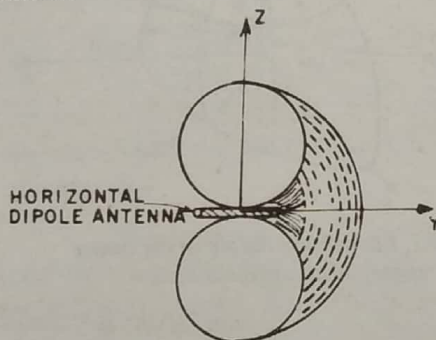


Fig. 6.3. (a). Half of the three dimensional pattern (Dipole placed horizontally).

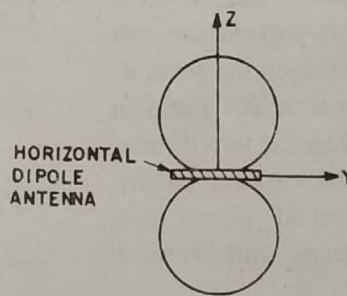


Fig. 6.3. (b) Two dimensional pattern obtained by cutting three dimensional pattern with a vertical plane along the axis of dipole (vertical pattern)

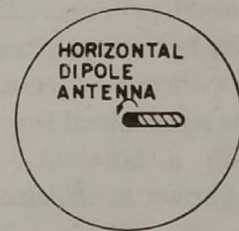


Fig. 6.3 (c) Two dimensional pattern when cut by a vertical plane at the centre of dipole perpendicular to its axis.

Since an isotropic radiator radiates uniformly in all direction and hence its radiation pattern is a sphere. The common types of radiation patterns can be classified as

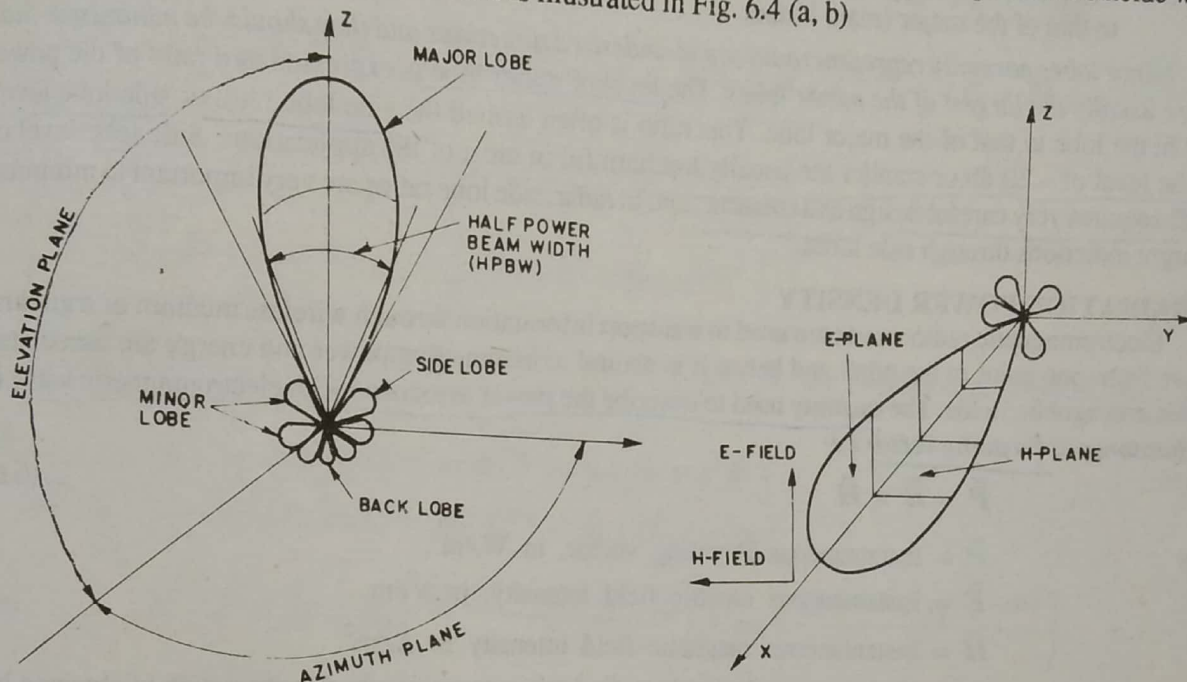
- (i) Omnidirectional or Broadcast-type pattern.
- (ii) Pencil Beam pattern.
- (iii) Fan-Beam Pattern.
- (iv) Shaped Beam Pattern.

In addition to these, there are other pattern shapes too which are utilized in direction finding work etc. For example, Limacon or Cardioid, figure of eight patterns etc.

On Summarising – An antenna radiation pattern is, therefore, defined as “a graphical representation of radiation properties of the antenna as a function of space co-ordinates. In most cases, the radiation pattern is determined in the far field region and is represented as a function of the directional coordinates. Radiation properties include radiation intensity field strength, phase or polarisation”. The radiation property of most concern is the three dimensional spatial distribution of radiated energy as a function

of the observer's position along a constant radius. A trace of the received power at a constant radius is called the "power pattern" and a graph of the spatial variation of the electric or magnetic field along a constant radius is called a "field pattern".

6.3.1. Principal Patterns. The performance of the antenna is usually described in terms of its principal E-plane and H-plane patterns. For a linearly polarised antenna, the E-plane pattern is defined as "the plane containing the electric-field vector and the direction of maximum radiation" and the H-plane as "the plane containing the magnetic-field vector and the direction of maximum radiation". It is normal practice to orient most antennas in such a way that atleast one of the principal plane patterns coincide with one of the "geometric principal planes. This is illustrated in Fig. 6.4 (a, b)



(a) Radiation lobes and BW of an antenna (b) E-plane and H-plane principal patterns..
Fig. 6.4.

In this, the $x-z$ plane is the principal E-plane (elevation plane $\phi = 0$) and the $x-y$ plane is the principal H-plane (azimuth plane $\theta = \pi/2$).

6.3.2. Radiation Pattern Lobes. Different parts of radiation pattern are referred to as "Lobes". This may be sub-classified as major lobe, minor lobe, side lobe and back lobe. A radiation lobe is a portion of the radiation pattern bounded by regions of relatively weak radiation intensity. This is illustrated in Fig. 6.5 where a symmetrical three-dimensional polar pattern with a number of radiation lobes. Although all are known as lobes but some are having greater radiation intensity and some are lesser radiation intensity. A linear two-dimensional pattern of the Fig. 6.4 (a) is shown in Fig. 6.5 where only one plane of Fig. 6.4 (a) is indicated.

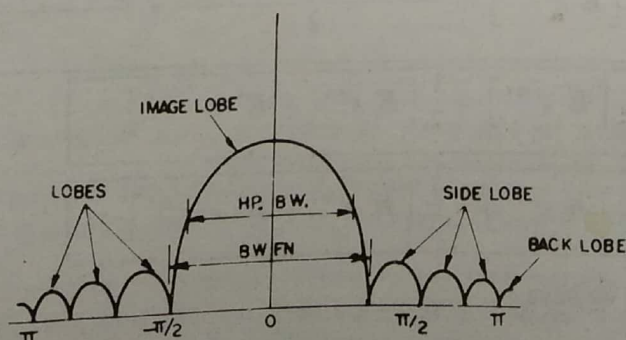


Fig. 6.5. Linear plot of power patterns.

- (i) **Major Lobe** is also called as main beam and is defined as “the radiation lobe containing the direction of maximum radiation”. In some antennas, there may exist more than one major lobe.
- (ii) **Minor Lobe** is any lobe except a major lobe *i.e.* all the lobes except the major lobes are called minor lobe.
- (iii) **Side Lobe** is a radiation lobe in any direction other than the intended lobe. Normally a side lobe is adjacent to the main lobe and occupies the hemisphere in direction of the main lobe.
- (iv) **Back Lobe** Normally refers to a minor lobe that occupies the hemisphere in a direction opposite to that of the major (main) lobe.

Minor lobes normally represent radiation in undesired directions and they should be minimised. Side lobes are usually the largest of the minor lobes. The level of minor lobe is expressed as a ratio of the power density in the lobe to that of the major lobe. This ratio is often termed the side lobe ratio or side lobe level. Side lobe level of - 20 db or smaller are usually not harmful in most of the applications. Side lobe level of - 30 db requires very careful design and construction. In radar, side lobe ratios are very important to minimise false target indications through side lobes.

6.4. RADIATION POWER DENSITY

Electromagnetic radio waves are used to transport information through wireless medium or a guiding structure from one point to the other and hence it is natural to assume that power and energy are associated with electromagnetic fields. The quantity used to describe the power associated with electromagnetic wave is the Instantaneous Poynting vector *i.e.*

$$\tilde{\mathbf{P}} = \tilde{\mathbf{E}} \times \tilde{\mathbf{H}} \quad \dots (6.6)$$

where

$\tilde{\mathbf{P}}$ = Instantaneous Poynting vector, in W/m².

$\tilde{\mathbf{E}}$ = Instantaneous electric field intensity, in V/m.

$\tilde{\mathbf{H}}$ = Instantaneous magnetic field intensity in A/m².

For time varying fields it is more desirable to find the average power density which is obtained by integrating the instantaneous Poynting vector over one period and dividing by the period.

For time harmonic variations of the form $e^{j\omega t}$, the complex fields $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are related by their instantaneous counterparts $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ as

$$\tilde{\mathbf{E}}_{(x, y, z, t)} = R_e \left[\mathbf{E}_{(x, y, z)} e^{j\omega t} \right] \quad \dots 6.7 (a)$$

$$\tilde{\mathbf{H}}_{(x, y, z, t)} = R_e \left[\mathbf{H}_{(x, y, z)} e^{j\omega t} \right] \quad \dots 6.7 (b)$$

The above equations can also be written as

$$R_e \left[\mathbf{E} e^{j\omega t} \right] = \frac{\mathbf{E} e^{j\omega t} + (\mathbf{E} e^{j\omega t})^*}{2} = \frac{\mathbf{E} e^{j\omega t} + \mathbf{E}^* e^{-j\omega t}}{2}$$

$$R_e \left[\mathbf{E} e^{j\omega t} \right] = \frac{1}{2} \left[\mathbf{E} e^{j\omega t} + \mathbf{E}^* e^{-j\omega t} \right] \quad \dots 6.8 (a)$$

Similarly

$$R_e \left[\mathbf{H} e^{j\omega t} \right] = \frac{1}{2} \left[\mathbf{H} e^{j\omega t} + \mathbf{H}^* e^{-j\omega t} \right] \quad \dots 6.8 (b)$$

Putting eqn. 6.7 into eqn. 6.6 we have

$$\tilde{P} = \tilde{E} \times \tilde{H} = R_e \left[E e^{j\omega t} \times H e^{j\omega t} \right]$$

... by eqn 6.6 & 6.7

$$= \frac{1}{2} \left[E e^{j\omega t} + E^* e^{-j\omega t} \right] \times \frac{1}{2} \left[H e^{j\omega t} + H^* e^{-j\omega t} \right]$$

... by eqn. 6.8

$$= \frac{1}{4} \left[(E \times H) e^{2j\omega t} + (E^* \times H) + (E \times H^*) + (E^* \times H^*) e^{-2j\omega t} \right]$$

$$= \frac{1}{4} \left[\{ (E^* \times H) + (E \times H^*) \} + \{ (E \times H) e^{2j\omega t} + (E^* \times H^*) e^{-2j\omega t} \} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} \{ (E^* \times H) + (E \times H^*) \} + \frac{1}{2} \{ (E \times H) e^{2j\omega t} + (E \times H) e^{+2j\omega t} \} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} \{ 2 R_e (E \times H^*) \} + \frac{1}{2} R_e \{ 2 R_e (E \times H) e^{2j\omega t} \} \right]$$

$$\tilde{P} = \frac{1}{2} \left[R_e (E \times H^*) \right] + \frac{1}{2} \left[R_e (E \times H) e^{2j\omega t} \right]$$

... (6.9)

since $(E^* \times H^*) e^{-2j\omega t} \equiv (E \times H) e^{+j\omega t}$. For, if E and H are expressed in complex polar and rectangular form, we may write

$$\tilde{E}(x, y, z) \equiv a_e E(x, y, z) = a_e E e^{j\theta_e} = a_e [E_r + j E_i]$$

and $\tilde{H}(x, y, z) \equiv a_h H(x, y, z) = a_h H e^{j\theta_h} = a_h [H_r + j H_i]$

Hence
$$\tilde{P} = \tilde{E} \times \tilde{H} = a_e \left[\frac{E e^{j\omega t} + E^* e^{-j\omega t}}{2} \right] \times a_h \left[\frac{H e^{j\omega t} + H^* e^{-j\omega t}}{2} \right]$$

$$= a_e \times a_h \times \frac{1}{4} \left[E H e^{j2\omega t} + E^* H^* e^{-2j\omega t} + E^* H + E H^* \right]$$

Then using the complex rectangular form we may write,

$$E H e^{j2\omega t} + E^* H^* e^{-2j\omega t} = 2 \left[(E_r H_r - E_i H_i) \cos 2\omega t - (E_i H_r + E_r H_i) \sin 2\omega t \right]$$

$$E H e^{j2\omega t} + E^* H^* e^{-2j\omega t} = 2 R_e \left[(E \times H) e^{2j\omega t} \right] \quad \dots (6.10)$$

and $E^* H = (E_r - j E_i) (H_r + j H_i) = E_r H_r + E_i H_i + j (E_r H_i - E_i H_r) \quad \dots 6.11 (a)$

and $E H^* = (E_r + j E_i) (H_r - j H_i) = E_r H_r + E_i H_i - j (E_r H_i - E_i H_r) \quad \dots 6.11 (b)$

Adding eqns. 6.11 (a) and 6.11 (b) we get

$$(E^* H + E H^*) = 2 (E_r H_r + E_i H_i) \quad \dots 6.12 (a)$$

or
$$(E^* H + E H^*) = 2 R_e [E \times H^*] = 2 R_e [E^* \times H] \quad \dots 6.12 (b)$$

Now it is evident from eqn. 6.9 that the first term is not a function of time and the time variations of the second term are twice the given frequency. Therefore, the average Poynting vector (or average power density) is written as

$$P_{av} = (\tilde{E} \times \tilde{H})_{av} = \frac{1}{2} R_e (E \times H^*) \quad \text{W/m}^2 \quad \dots (6.13)$$

where E and H are peak value and the term $\frac{1}{2}$ is omitted when converted to RMS value. Hence the real part of eqn. 6.9 represents the average (real) power density and the imaginary part of the same quantity represent the reactive (stored) power density associated with the electromagnetic fields. The power density associated with the electromagnetic fields of an antenna in its far field region is predominantly real which is referred to as radiation density. Further the average power radiated by an antenna (radiated power) is given by

$$P_{\text{rad.}} = P_{\text{av}} = \oint_S P_{\text{rad.}} \cdot ds = \oint P_{\text{av}} \cdot ds = \frac{1}{2} \oint_S R_e (E \times H^*) \cdot ds \quad \dots (6.14)$$

6.5. RADIAN AND STERADIAN

In order to measure a plane angle radian is used. One radian is defined as the plane angle with its vertex at the centre of a circle of radius r that is subtended by an arc whose length is r , as shown in Fig. 6.6 (a). As the circumference of a circle radius r is $c = 2\pi r$, there are 2π radian i.e. $\left(\frac{2\pi r}{r}\right)$ in a full circle.

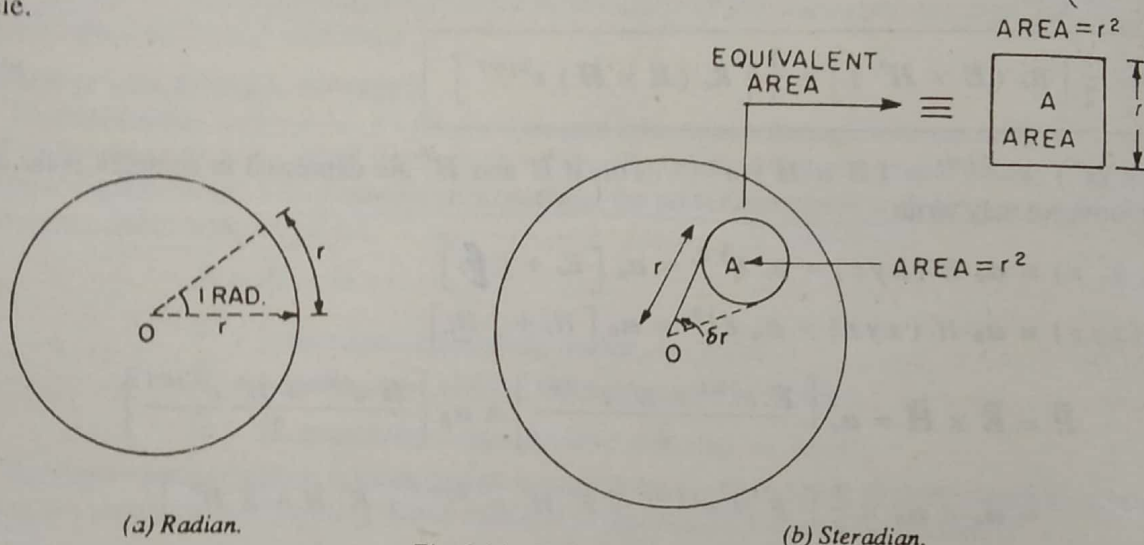


Fig. 6.6 (a, b). Radian and Steradian.

In a similar way the measure of a solid angle is a steradian. One steradian is defined as the solid angle with its vertex at the centre of a sphere of radius r i.e. subtended by a spherical surface area equal to that of a square with each side of length r . This is illustrated in Fig. 6.6 (b). As the area of a sphere of radius r is $A = 4\pi r^2$, there are 4π steradian i.e. $\left(\frac{4\pi r^2}{r^2}\right)$ in a closed sphere. The infinitesimal area ds on the surface of a sphere of radius r is

$$ds = r^2 \sin \theta d\theta d\phi \quad \text{m}^2 \quad \dots 6.15 (a)$$

so the element of solid angle $d\Omega$ of a sphere is written as

$$d\Omega = \frac{ds}{r^2} = \sin \theta d\theta d\phi \quad (\text{Sr}) \quad \dots 6.15 (b)$$

6.6. RADIATION INTENSITY

(AMIETE, Nov. 1976, May 1976, 1978)

Radiation intensity is a quantity which does not depend upon the distance from the radiator and is denoted by a letter capital U or Φ . It helps in defining various antenna terms like Antenna gain or simply gain or directive gain or power gain or directivity. Radiation intensity is defined as "Power per unit solid angle". Since unit of power and solid

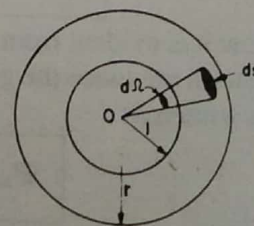


Fig. 6.7. Illustration of Radiation Intensity.

angle in S.I. units are 'Watts' and 'Steradian' respectively and hence unit of radiation intensity is Watts/steradian or Watts/radian².

In order to express radiation intensity mathematically, let us consider the Fig. 6.7.

It is clear from Fig. 6.7 that solid angle $d\Omega$ is given by solid angle definition explained above.

$$\boxed{d\Omega = \frac{ds}{r^2}} \quad \dots 6.16 (a)$$

or $ds = r^2 d\Omega$ ds = elemental surface area
 $d\Omega$ = differential solid angle.

$$\boxed{\frac{ds}{d\Omega} = r^2} \quad \dots 6.16 (b)$$

Thus there are r^2 metres of surface area per unit solid angle Eqn. 6.12 (b). Further it is known that the power radiated per unit area in any direction is given by the Poynting vector.

$$\mathbf{P} = \mathbf{E} \times \mathbf{H} \left(\frac{\text{Volt}}{\text{m}} \times \frac{\text{Amp}}{\text{m}} = \text{Watts/m}^2 \right)$$

where

$\mathbf{E}$ = Electric field Intensity, in volts/m.

$\mathbf{H}$ = Magnetic field Intensity, in A/m.

$\mathbf{P}$ = Instantaneous Poynting vector, in Watts/m².

Here $\mathbf{E} \times \mathbf{H}$ is a vector product and the mutually perpendicular components $\mathbf{E}$ and $\mathbf{H}$ contribute anything to the power flow and the direction of the flow is normal to the plane containing $\mathbf{E}$ and $\mathbf{H}$. But when $\mathbf{E}$ and $\mathbf{H}$ are changing with time then average Poynting vector ($\mathbf{P}_{av}$), instead of simple $\mathbf{P}$, is of great interest. In order to find ($\mathbf{P}_{av}$) complex notation has to be followed. In complex notation Poynting vector is given by

$$\mathbf{P} = \frac{1}{2} (\mathbf{E} \times \mathbf{H}^*) \quad \left(\because W = \frac{1}{2} VI \right)$$

where

$$\mathbf{E} = \mathbf{a}_e E_0 e^{j\omega t} \quad \dots 6.14 (a)$$

and

$\mathbf{H}^*$ = complex conjugate of $\mathbf{H}$.

$$\mathbf{H}^* = \mathbf{a}_h H_0 e^{-j(\omega t - \delta)} \quad \dots 6.14 (b)$$

Here $\mathbf{a}_e$ and $\mathbf{a}_h$ are unit vectors in the direction of $\mathbf{E}$ and $\mathbf{H}$ respectively. Also

$$\mathbf{H} = \mathbf{a}_h H_0 e^{j(\omega t - \delta)}$$

and

δ = Phase angle between $\mathbf{E}$ and $\mathbf{H}$.

Now we are in position to give average Poynting vector ($\mathbf{P}_{av}$) which is given by the real part of the complex Poynting vector i.e.

$$\mathbf{P}_{av} = \text{Real part of } \mathbf{P} \text{ (complex)} = R_e \mathbf{P}$$

$$\mathbf{P}_{av} = \frac{1}{2} R_e \{ \mathbf{E} \times \mathbf{H}^* \} \text{ Watts} \quad \dots (6.13)$$

In this $\mathbf{E}$ and $\mathbf{H}$ are instantaneous values. If r.m.s. values are taken then the term $\left(\frac{1}{2} \right)$ will be omitted. Alternatively, $\mathbf{P}_{av}$ can also be found by integrating the instantaneous Poynting vector over one period and dividing the same by one period.

It is known for radiation field that E and H are orthogonal in a plane normal to radius vector and for this case.

$$E = 120 \pi H \quad \dots 6.15 (a)$$

$$E = \eta_0 H \quad \dots 6.15 (b)$$

where

η_0 = Intrinsic Impedance of space.

Thus power flow per unit area can now be written as

$$P = EH = E \frac{E}{\eta_0} \quad \text{from above}$$

or

$$P = \frac{E^2}{\eta_0} \quad \text{W/m}^2 \quad \dots 6.16 (a)$$

In the far field, Poynting vector is entirely radial i.e. $|P| = P_r$. Fields are entirely transverse and E and H are varying as $1/r$ so that, radiation intensity is independent of distance. Instead radiation Intensity depends on the angle θ and Φ . Therefore, the relation between average Poynting vector and electric field at a point of the far field is given by

$$P_r = \frac{1}{2} \frac{E^2(\theta, \phi)}{\eta_0} \quad \dots 6.16 (b)$$

If now the magnitude of Poynting vector P_r is multiplied by the square of radius r at which radiation intensity is measured, then we get the power per unit solid angle i.e. the radiation Intensity.

$$\therefore \text{Radiation Intensity} \equiv \Phi(\theta, \phi) = P_r \cdot r^2 \equiv \Phi$$

or

$$\Phi = P_r \cdot r^2 \quad \dots 6.17 (a)$$

or

$$\Phi = \frac{1}{2} \frac{E^2(\theta, \phi)}{\eta_0} r^2 \quad \text{from eqn. 6.16 (b)} \quad \dots 6.17 (b)$$

The total power radiated (W_r) from a radiator is calculated from

$$\Phi = \frac{\text{Differential Power } (dW_r)}{\text{Differential element of solid angle } (d\Omega)}$$

or

$$dW_r = \Phi d\Omega \quad \dots (6.18)$$

Total power radiated is obtained by integrating eqn. 6.18

$$\text{i.e.} \quad W_r = \int dW_r = \int d\Omega \cdot \Phi$$

or

$$W_r = \int \Phi d\Omega \quad \text{Watts} \quad \dots (6.19)$$

Also for a isotropic radiator, total solid angle contains 4π steradians

$$\therefore \quad W_r = \int \Phi \cdot 4\pi \text{ Watts} = 4\pi \int \Phi \quad \Phi = \Phi_{av} \text{ (say)}$$

$$W_r = 4\pi \cdot \Phi_{av} \quad \dots 6.20 (a)$$

or

$$\Phi_{av} = \frac{W_r}{4\pi} \quad \text{W/Sr} \quad \dots 6.20 (b)$$

where

Φ_{av} = Radiation Intensity produced by an isotropic radiator, radiating the same total power W_r .

Since radiation intensity $\Phi(\theta, \phi)$ depends upon the angle θ and ϕ therefore, it can be represented in any from as

(3) In terms of power input, power gain is given by

$$\text{Power gain } (G_p) = \frac{\text{Power input supplied to subject antenna in the direction of maximum radiation}}{\text{Power input supplied to reference antenna}} \dots (6.34)$$

for the same field strength at the same given point.

For example, let the field produced by a subject antenna is twice to that produced by an isotropic antenna, i.e.

$$G_0 = \frac{v_1}{v_2} = 2$$

$$\therefore \text{Power gain } G_p = \left(\frac{v_1}{v_2} \right)^2 = 2^2 = 4$$

This shows that to produce the same field strength at the same distance, four times as much power would have to be supplied to an isotropic radiator in comparison to the practical antenna.

It may, however, be noted that in gain or power gain antenna efficiency is involved and an account is taken **for the input power** antenna losses, and power consumed in terminating resistance. The power gain depends on the following two things :

(i) **Sharpness of lobe.** Sharper the lobe higher will be the power gain.

(ii) **Volume of the solid radiation pattern.** Since decibel is the excellent practical unit for measuring power ratio and hence power gain of antenna system is generally expressed in decibels. The number of decibels corresponding to any power ratio is equal to 10 times the common logarithm of the power ratio, i.e.

$$G_p (db) = 10 \log_{10} G_p$$

$$G_p (db) = 10 \log_{10} \frac{P_1}{P_2} \dots 6.35 (a)$$

$$\text{or } G_p (db) = 10 \log_{10} \left(\frac{v_1}{v_2} \right)^2 = 10 \times 2 \log_{10} \left(\frac{v_1}{v_2} \right) \quad \text{if voltage ratio is involved}$$

$$\text{or } G_p (db) = 20 \log_{10} \left(\frac{v_1}{v_2} \right) \dots 6.35 (b)$$

6.9.1. Practical Importance of Power Gain. For a given amount of input power to an antenna, the power density at a given point in the space is proportional to the power gain of the antenna in that direction. Hence the signal available to a receiving antenna at that location can be increased by increasing the transmitted power without increasing the transmitted power. For example, transmitter with a power out of 1000 W and an antenna with a power gain of 10 (i.e. $G_{db} = 10 \log_{10} 10$ or $G_{db} = 10 \times 1.000 = 10 \text{ db}$) will give the same power density at a receiving point as will be a transmitter of 500 W power output and an antenna of power gain of 20 (i.e. $10 \log_{10} 20 = 10 \times 1.3010 \approx 13.0 \text{ db}$). *Therefore this relationship is of great economic importance, obviously.* Sometimes it may be much less expensive to double the gain of the antenna (and add 3 db, rather than to double the transmitted power (Although converse may also be true at other times). However, it is desirable to use as much antenna gain as may possibly be obtained, when it is desirable to give the maximum possible field strength in a particular direction.

Sometimes when it is desirable to radiate equally in all horizontal directions, then maximum gain obtainable is limited due to the connection between antenna gain and antenna pattern. Thus determination of gain which is required in a given application is a matter of engineering judgement.

6.10. DIRECTIVITY (D)

(AMIETE, May 1974, 1976)

The maximum directive gain is called as directivity of an antenna and is denoted by D . In a particular direction the directivity D is a constant.

(1) Directivity of an antenna is defined as the ratio of Maximum radiation intensity to its average radiation intensity i.e.

$$\text{Directivity } (D) = \frac{\text{Maximum Radiation Intensity of test antenna}}{\text{Average Radiation Intensity of test antenna}}$$

or
$$D = \frac{\Phi(\theta, \varphi)_{\max}}{\Phi_{av}} \text{ both of test antenna.} \quad \dots(6.36)$$

(2) Directivity D of an antenna may also be defined as the ratio of maximum radiation intensity of the subject antenna to the radiation intensity of an isotropic or reference antenna radiating the same total power i.e.

$$\text{Directivity} = \frac{\text{Maximum Radiation Intensity of subject or test antenna}}{\text{Radiation Intensity of an isotropic antenna}}$$

or
$$D = \frac{\Phi(\theta, \varphi)_{\max} \cdot (\text{test antenna})}{\Phi_0 (\text{isotropic antenna})} \quad \dots(6.37)$$

(3) Alternatively still another way of expressing Directivity of an antenna is in terms of total radiated power. It is ratio of total radiated power by the subject antenna to the power radiated by an isotropic antenna for the same radiation intensity, i.e.

$$\text{Directivity} = \frac{\text{Power radiated from a test antenna}}{\text{Power radiated from an isotropic antenna}}$$

For the same radiation intensity

or
$$D = \frac{W'}{W''} \frac{(\text{from a test antenna})}{(\text{from an isotropic antenna})} \quad \dots(6.38)$$

Since the average radiation intensity (Φ_{av}) is obtained by dividing total power radiated W by 4π steradian. Therefore eqn. 6.30 may be written as

$$D = \frac{\Phi(\theta, \varphi)_{\max}}{W/4\pi}$$

$$D = \frac{4\pi \Phi(\theta, \varphi)_{\max}}{W}$$

... 6.39 (a)

or

$$D = \frac{4\pi (\text{maximum radiation intensity})}{\text{Total radiated power}}$$

... 6.39 (b)

The directivity implies the maximum values for an antenna. The directivity D for which radiation intensity Φ is not maximum is designated by specifying an angle φ at which the directivity is measured.

In general directivity may be represented by $D(\theta, \varphi)$ and is given by

$$D(\theta, \varphi) = \frac{\Phi_{\max}}{\Phi} = D$$

or

$$D(\theta, \varphi) = \frac{\Phi}{\Phi_{\max}} D$$

... (6.40)

where $\Phi_{\max}$ = max. radiation intensity.

Φ = Radiation intensity in direction of (θ, φ) .

The directivity D is also expressed in decibels ratio as

$$(\text{Directivity}) \text{ db} = 10 \log_{10} D \quad \dots 6.41 (a)$$

$$D (\text{db}) = 10 \log_{10} D \quad \dots 6.41 (b)$$

or

The numerical value of directivity D always lies between 1 and ∞ i.e. $1 \leq D \leq \infty$. 1 being the directivity of an isotropic antenna and hence D can not be less than 1.

Let us now find a **general expression** for the directivity. For, the radiation intensity may be any function of θ and ϕ and hence, in general, represented as

$$\Phi = \Phi_a f(\theta, \phi) \quad \dots (6.42)$$

where

$$\Phi_a = \text{a constant} \quad \Phi = \text{Radiation intensity}$$

$\therefore$ Maximum value of Radiation intensity (eqn. 6.36) is given by

$$\Phi_{\max} = \Phi_a f(\theta, \phi)_{\max} \quad \dots (6.43)$$

$$\text{As a special case, in case of anisotropic antenna } f(\theta, \phi)_{\max} = 1 \quad \dots (6.44)$$

$$\text{Then } \Phi_{\max} = \Phi_a \quad \dots (6.45)$$

Putting eqn. 6.45, in eqn. 6.42, we get

$$\Phi = \Phi_m f(\theta, \phi) \quad \dots (6.46)$$

Further, the average radiation intensity Φ_{av} is given by total power radiated divided by 4π steradian.

$$\begin{aligned} \Phi_{av} &= \frac{W_r}{4\pi} = \frac{\iint \Phi d\Omega}{4\pi} \\ \Phi_{av} &= \frac{\iint \Phi_a f(\theta, \phi) d\Omega}{4\pi} \end{aligned} \quad \dots 6.46 (a)$$

where

$$d\Omega = \sin \theta d\theta d\phi = \text{elements of solid angle.}$$

Thus by definition, directivity

$$\begin{aligned} D &= \frac{\text{Max. R.I. } (\Phi_{\max})}{\text{Average R.I. } (\Phi_{av})} = \frac{\Phi_a f(\theta, \phi)_{\max}}{\iint \frac{\Phi_a f(\theta, \phi) d\Omega}{4\pi}} \\ &= \frac{4\pi f(\theta, \phi)_{\max}}{\iint f(\theta, \phi) d\Omega} \quad \dots (6.47) \\ &= \frac{4\pi}{\iint f(\theta, \phi) d\Omega} = \frac{4\pi}{\iint f_n(\theta, \phi) d\Omega} \end{aligned}$$

from eqn. 6.42

$$\therefore \frac{f(\theta, \phi)}{f(\theta_{\max}, \phi)} = f_n(\theta, \phi) = \text{normalized power pattern}$$

$$D = \frac{4\pi}{B.A.} \quad \dots 6.48 (a)$$

where

$$B.A. = \frac{\iint f(\theta, \phi) d\Omega}{f(\theta, \phi)_{\max}} = \text{Beam Area} \quad \dots 6.48 (b)$$

$$D = \frac{4\pi}{\Omega_A} \quad \dots 6.48 (c)$$

where

$$\Omega_A = \text{Beam solid angle}$$

$$\Omega_A = \frac{\iint f(\theta, \phi) d\Omega}{f(\theta, \phi)_{\max}} \quad \dots (6.49)$$

Therefore, the directivity of an antenna is nothing but solid angle of a sphere (i.e. 4π steradian) divided by the antenna beam solid angle Ω_A .

From the definition of directivity eqn. 6.36 and eqn. 6.48

$$D = \frac{\Phi_{\max}}{\Phi_{av}} = \frac{4\pi}{\Omega_A}$$

or $4\pi \Phi_{av} = \Omega_A \Phi_{\max} \quad \therefore \Phi_{av} = \frac{W_r}{4\pi}$

or $4\pi \cdot \frac{W_r}{4\pi} = \Omega_A \Phi_{\max}$

$$W_r = \Omega_A \Phi_{\max} \quad \dots (6.50)$$

where W_r = Total power radiated.

Thus beam solid angle is the solid angle through which all the power radiated would stream if the power per unit solid angle equal to the maximum value of radiation intensity $\Phi_{\max}$ over the beam solid angle.

Directivity is a dimensionless (i.e. constant) quantity which indicates the effectiveness of concentrating power into a limited solid angle. The narrower the solid angle, the higher the directivity.

It entirely depends on the far field pattern. Unlike gain or power gain, antenna efficiency is not involved in the directivity. If an antenna has not any losses like ohmic, dielectric mismatch i.e. 100% efficient, then directivity and gain are same. However, for an antenna with losses, gain will be less than directivity by a factor which corresponds to efficiency. The directivity and gain are related as

$$G_0 = kD \quad \dots (6.51)$$

where

G = gain

k = efficiency factor = 1 for 100% efficiency

< 1 if losses are present

D = Directivity

This is derived as follows. Let us assume that the maximum radiation intensity from the test antenna is $\Phi_{\max}$ and let this maximum radiation intensity $\Phi'_{\max}$ be related to the maximum radiation intensity $\Phi_{\max}$ of 100% efficient test antenna by a factor k known as radiation efficiency factor as

$$\Phi'_{\max} = k \Phi_{\max} \quad \dots (6.52)$$

where the values of k lies between 0 and 1 i.e. $0 \leq k \leq 1$

But the gain w.r.t. an isotropic antenna is

$$G_0 = \frac{\Phi'_{\max}}{\Phi_0} \quad \text{from Eqn. 6.24 (b)}$$

where Φ_0 = R.I. from a lossless isotropic antenna

or $G_0 = \frac{k \Phi_{\max}}{\Phi_0} \quad \text{from Eqn. 6.52}$

From Eqn. 6.37

$$G_0 = kD \quad \dots 6.52 (a)$$

$$D = \frac{\Phi_{\max}}{\Phi_0}$$

Since, in many antennas the antenna losses are extremely small and hence the value of gain is almost equal to the directivity. That is why, gain and directivity are interchangeably used.

6.11. DIRECTIVE GAIN AND GAIN OR POWER GAIN

(1) By definition, directive gain G_d is given by

$$\text{Directive gain } G_d = \frac{\text{Radiation Intensity}}{\text{Average Radiated Power}} = \frac{\Phi(\theta, \phi)}{W_r/4\pi}$$

or

$$G_d = \frac{4\pi\Phi(\theta, \phi)}{W_r} \quad \dots (6.53)$$

But the power gain is the ratio of two powers and the total power input is taken instead of simply radiated power i.e.

$$\text{Power gain or gain} = \frac{\text{Radiation intensity}}{\text{Total input power}} G_p = \frac{\Phi(\theta, \phi)}{W_T/4\pi}$$

or

$$G_p = \frac{4\pi\Phi(\theta, \phi)}{W_T} \quad \dots (6.54)$$

where

$$W_T = W_r + W_l$$

Total Power = Radiated Power + Power loss in ohmic resistance

(2) Directive gain depends entirely on the distribution of radiated power in space whereas gain or power gain depends on the sharpness of the lobe and volume of the solid radiation pattern.

(3) Directive gain is a qualitative measure of the extent to the total power radiated is concentrated in one direction and it does not depend upon the power input to the antenna and antenna losses like dielectric losses, Cu losses, mismatch losses and power consumed in terminating resistance etc. whereas the power input and the antenna losses are accounted in the power gain. Thus power gain takes into account the antenna efficiency as well as its directional properties.

6.12. DIRECTIVE GAIN AND DIRECTIVITY

(AMIETE, May 1978, Dec. 1973)

(1) By definition,

$$\text{Directive gain} = \frac{\text{Radiation Intensity}}{\text{Average radiated power}}$$

or

$$G_d = \frac{\Phi(\theta, \phi)}{W_r/4\pi} = \frac{4\pi\Phi(\theta, \phi)}{W_r} \quad \dots (6.53)$$

The extent to which a practical antenna concentrates its radiated energy relative to that of some standard antenna is known as directive gain whereas the maximum directive gain is called as directivity D and is defined as

$$\text{Directivity } (D) = \frac{\text{Maximum radiation intensity of a test antenna}}{\text{Average radiation intensity of test antenna}}$$

or

$$D = \frac{\Phi(\theta, \phi)_{\max}(\text{test antenna})}{\Phi_{av}(\text{test antenna})}$$

Alternatively,

$$\text{Directivity} = \frac{\text{Maximum Radiation Intensity of test antenna}}{\text{Radiation Intensity of an isotropic antenna}}$$

or

$$D = \frac{\Phi(\theta, \phi)_{\max}(\text{test antenna})}{\Phi_0(\text{isotropic antenna})} \quad \dots (6.37)$$

(2) For a lossless isotropic antenna, directive gain and directivity is same.

In this case radiation efficiency factor $k = 1$.

$\therefore$

$$G_0 = kD$$

or

$$G_0 = D \quad \text{if } k = 1$$

(3) The numerical value of directive gain may lie between 0 and ∞ whereas that of directivity between 1 and ∞ and in case of directivity it can not be less than 1.

(4) Directive gain depends entirely on the distribution of radiated power in space whereas the value of directivity depends on the solid angle of the far (radiated) field pattern.

(5) Directive gain does not depend on the power input to the antenna and antenna losses and so is true for the directivity. In other words antenna efficiency is not involved in these two.

(6) The numerical value of directivity of current element and half-wave dipole is respectively 1.5 (or 1.76 db) and 1.64 (or 2.15 db).

$\therefore$ Directive gain of half-wave dipole over current element = $(2.15 - 1.76) \text{ db} = 0.39 \text{ db}$.

6.13. ANTENNA EFFICIENCY (η)

(Kanpur Univ. M.Sc. Phy. (Prev.) 1981, (AMIETE, May 1968, Dec. 1973)

The efficiency of an antenna is defined as the ratio of power radiated to the total input power supplied to the antenna and is denoted by η or k . Thus

$$\text{Antenna efficiency} = \frac{\text{Power Radiated}}{\text{Total Input Power}}$$

$$\begin{aligned} \text{or } \eta &= \frac{W_r}{W_T} = \frac{W_r}{W_r + W_l} = \frac{W_r}{W_T} \times \frac{4\pi\Phi(\theta, \varphi)}{4\pi\Phi(\theta, \varphi)} \\ &= \frac{4\pi\Phi(\theta, \varphi)}{W_T} \cdot \frac{W_r}{4\pi\Phi(\theta, \varphi)} = G_p \cdot \frac{1}{G_d} = \frac{G_p}{G_d} \end{aligned} \quad \begin{array}{l} \text{From Eqn. 6.33 (b)} \\ \text{and 6.30 (a)} \end{array}$$

$$\text{or } \boxed{\eta = \frac{G_p}{G_d} = \frac{W_r}{W_r + W_l}} \quad \dots (6.55)$$

where

W_r = Power radiated.

W_l = Ohmic losses.

If current flowing in the antenna is I , then $\eta = \frac{I^2 R_r}{I^2 (R_r + R_l)}$

$$\text{or } \boxed{\eta\% = \frac{R_r}{R_r + R_l} \times 100} \quad \dots (6.56)$$

where R_r = Radiation resistance ; R_l = Ohmic loss resistance of antenna conductor and $R_r + R_l$ = Total effective resistance.

It is desirable to have a better radiation characteristics from the antenna and for this loss resistances should be as small as possible. The loss resistances may consist of the following, in general.

- (i) Ohmic loss in the antenna conductor.
- (ii) Dielectric loss.
- (iii) $I^2 R$ loss in antenna and ground system.
- (iv) Loss in earth connections.
- (v) Leakage loss in insulation.

Thus antenna efficiency η represents the fraction of total energy supplied to the antenna which is converted into electromagnetic waves.

6.14. EFFECTIVE AREA OR EFFECTIVE APERTURE OR CAPTURE AREA

A transmitting antenna transmits electromagnetic waves and a receiving antenna receives a fraction of the same. The concept of effective area or aperture is best understood by considering an antenna to have an

(AMIETE, Nov. 1977, 87)

effective area or aperture over which it extracts electromagnetic energy from the travelling electromagnetic waves. It may be defined as the ratio of power received at the antenna load terminal to the Poynting vector (or power density) in Watts/metre² of the incident wave. Thus

$$\text{Effective area or Effective aperture or Capture area} = \frac{\text{Power received}}{\text{Poynting vector of incident wave}} \quad \dots 6.57 (a)$$

$$\text{or} \quad A_e = \frac{W}{P} \equiv A$$

$$\text{or} \quad \boxed{W = PA} \quad \dots 6.57 ((b))$$

where

W = Power received, in Watts.

P = Poynting vector of incident plane wave, in Watts/m² or power flow per sq. metre for the incident wave.

A = Effective or Capture area or Effective aperture, in m².

Let a receiving antenna be placed in the field of plane polarised travelling waves as illustrated in Fig. 6.7 having an effective area A and the receiving antenna (dipole) is terminated at a load impedance $Z_L = R_L + jX_L$.

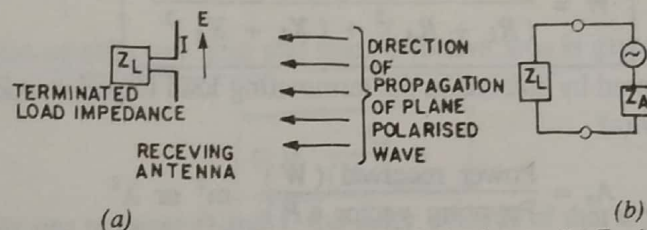


Fig. 6.7 (a). Receiving antenna in the field of plane polarised wave. (b) Equivalent circuit of Fig. (a).

If I be the terminal current, then received power

$$W = I_{rms}^2 R_L \quad \dots (6.58)$$

where

R_L = Load resistance, in Ω

I_{rms} = Terminal rms current

$$\therefore A = \frac{W}{P} = \frac{I_{rms}^2 R_L}{P}$$

Since the antenna extracts energy from incident electromagnetic waves, delivers the same to terminated load impedance Z_L and the power flowing per square metre or Poynting vector is P W/m². This entire system can be replaced by an equivalent circuit [given in Fig. 6.7 (b)] according to Thevenin's theorem. In Fig. 6.7(b)

V = Equivalent Thevenin's voltage.

Z_A = Equivalent Thevenin's impedance.

The voltage V is induced by passing electromagnetic waves which produces current I_{rms} through terminal load impedance Z_L .

$$\therefore I_{rms} = \frac{\text{Equivalent voltage}}{\text{Equivalent Impedance}}$$

$$I_{rms} = \frac{V}{Z_L + Z_A} \text{ Amp.} \quad \dots (6.59)$$

where

$Z_A = R_A + jX_A$ = complex antenna impedance.

and $R_A = R_r + R_l = R_r$ if $R_l = 0$ is assumed (6.60)
 = Radiation resistance + loss resistance

Now putting the values of Z_L and Z_A , we have

$$I_{rms} = \frac{V}{(R_L + jX_L) + (R_A + jX_A)}$$

or $|I_{rms}| = \frac{|V|}{\sqrt{(R_L + R_A)^2 + (X_L + X_A)^2}}$ 6.61 (a)

or $|I_{rms}| = \frac{|V|}{\sqrt{(R_L + R_r + R_l)^2 + (X_L + X_A)^2}}$ 6.61 (b)

where

X_L = Load reactance, in Ω

X_A = Antenna reactance, in Ω

Since the Power received by terminal load impedance is given by eqn. 6.58. Therefore, putting eqn. 6.61 (a) in 6.58, we get,

$$W = I_{rms}^2 \cdot R_L$$

$$W = \frac{V^2 R_L}{(R_L + R_A)^2 + (X_L + X_A)^2} \quad \dots (6.62)$$

This is the power delivered by antenna at the terminating load impedance Z_L . Now by definition of effective Aperture A_e , eqn. 6.57 (a)

$$A_e = \frac{\text{Power received } (W)}{\text{Poynting vector } (P)} \quad \text{m}^2 \text{ or } \lambda^2$$

$$= \frac{V^2 R_L}{[(R_L + R_A) + (X_L + X_A)^2] P} \quad \text{m}^2 \text{ or } \lambda^2 \quad \text{From eqn. 6.62}$$

Since W is in Watts and P in Watts/metre² and hence the unit of A_e is m². But if the unit of P is taken in terms of wavelength λ (which is convenient) as Watts/(wave length)², then A_e is in (wavelength)²

$$\therefore A_e = \frac{V^2 R_L}{[(R_L + R_A)^2 + (X_L + X_A)^2] P} \quad \lambda^2 \quad \dots 6.63 (a)$$

$$A_e = \frac{V^2 R_L}{[(R_L + R_r + R_l)^2 + (X_L + X_A)^2] P} \quad \lambda^2 \quad \dots 6.63 (b)$$

This is the expression for effective area or aperture or capture area in terms of power density (or Poynting vector P), induced voltage V , antenna impedance Z_A and terminating load impedance Z_L . The induced voltage V is maximum when antenna is oriented for max. response and the antenna and the incident wave both have the same polarisation. In eqn. 6.63, $R_A = R_r + R_l$ where R_l accounts for any mismatch between Antenna and Z_L and antenna losses.

According to maximum power transfer theorem, maximum power will be transferred from antenna to the antenna terminating load if

and

or

$$X_L = -X_A$$

$$R_L = R_A = R_r + R_l$$

$$R_L = R_r \quad \text{if } R_l = 0$$

$$\dots 6.64 (a)$$

$$\dots 6.64 (b)$$

Thus maximum power received in Antenna terminating load impedance Z_L can be obtained if conditions of eqn. 6.64 is substituted in eqn. 6.62

$$W_{\max} = \frac{V^2 R_L}{4 R_L^2} = \frac{V^2}{4 R_L} = \frac{V^2}{4 R_r}$$

or

$$W_{\max} = \frac{V^2}{4 R_r}$$

$$\therefore R_L = R_r \quad \dots (6.65)$$

This is the maximum power received in antenna terminating load impedance Z_L under the condition of max. power transfer and without antenna loss and corresponding effective aperture is known as maximum effective aperture.

$\therefore$ Maximum Effective Aperture

$$(A_e)_{\max} = \frac{\text{Max. recd. power}}{\text{Power density of incident wave}} = \frac{V^2}{4 R_r P}$$

$$(A_e)_{\max} = \frac{V^2}{4 P R_r} \quad \lambda^2 \text{ or } m^2$$

$$\dots (6.66)$$

Sometimes max. effective aperture is simply designated as effective aperture and effective aperture as simply aperture following the 'Institute of Radio Engineers' (IRE) standards. Eqn. 6.66 is the expression for the maximum Effective Aperture in terms of induced voltage, power density of incident wave and radiation resistance of antenna.

Further, the ratio of effective area and max. effective area is given a name known as effectiveness ratio and is denoted by α .

Thus

$$\alpha = \frac{A_e}{(A_e)_{\max}}$$

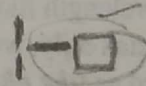
$$\dots (6.67)$$

The values of α lies between 0 and 1, the latter value is of that a perfectly matched antenna of cent per cent efficiency.

In practice, however, the terminated load impedance is generally, not at the antenna terminal as shown in Fig. 6.7 but the load is in the receiver which itself is connected with a transmission line. In such case the load impedance Z_L is the equivalent impedance which appears across the terminals of the antenna. Thus the power delivered to receiver and load impedance Z_L is same if transmission line is lossless and less if the T/L is lossy by an amount equal to loss in the line.

6.14.1. Scattering loss Apertures. Besides effective aperture, there are other apertures also like scattering aperture (A_s) and loss aperture (A_l) corresponding to considerable losses in radiation or re-radiation resistance (R_r) and antenna loss resistance (R_l) respectively and accordingly they are called as scattering and loss apertures. Mathematically,

$$\text{Scattering Aperture} = A_s = \frac{I_{rms}^2 R_r}{P}$$



or

$$A_s = \frac{V^2 R_r}{[(R_L + R_A)^2 + (X_L + X_A)^2] P}$$

$$\dots (6.68)$$

and

$$\text{Loss Aperture} = A_l = I_{rms}^2 R_l$$

or

$$A_l = \frac{V^2 R_l}{[(R_L + R_A)^2 + (X_L + X_A)^2] P}$$

$$\dots (6.69)$$

If now the condition of maximum transfer of energy is introduced in eqn. 6.68 from eqn. 6.64, we get

$$(A_s)_{\max} = \frac{V^2 R_r}{(R_L + R_r)^2 P} = \frac{V^2 R_r}{4 R_r^2 P} \quad \because R_A = R_r + R_l = R_r \text{ if } R_l = 0 \text{ but } R_L = R_r$$

$$(A_s)_{\max} = \frac{V^2}{4 R_r P} \quad \dots (6.70)$$

Thus under the condition of maximum transfer of energy maximum scattering aperture and maximum effective aperture are same as seen from eqn. 6.66 and 6.70 i.e.

$$(A_s)_{\max} = (A_e)_{\max} \quad \dots (6.71)$$

Further, the ratio of scattering aperture to effective aperture is given a name known as scattering ratio and is denoted by β . Thus

$$\text{Scattering ratio} = \beta = \frac{A_s}{A_e}$$

The values of β lies between 0 to ∞ .

6.14.2. Collecting Aperture. Out of power collected by antenna, there are losses as heat in load resistance (R_l), radiation resistance (R_r) and antenna loss resistance (R_L) and correspondingly the three apertures are Effective, Scattering and Loss. By virtue of conservation of energy these three apertures are collectively known as **collecting aperture** and is given mathematically as

$$\text{Collecting Aperture} = A_c = A_e + A_s + A_l$$

$$\begin{aligned} \text{or} \quad A_c &= I_{rms}^2 R_L + I_{rms}^2 R_r + I_{rms}^2 R_l \\ &= \frac{V^2 R_L}{P [(R_L + R_A)^2 + (X_L + X_A)^2]} + \frac{V^2 R_r}{P [(R_L + R_A)^2 + (X_L + X_A)^2]} \\ &\quad + \frac{V^2 R_l}{P [(R_L + R_A)^2 + (X_L + X_A)^2]} \end{aligned}$$

$$A_c = \frac{V^2 (R_L + R_r + R_l)}{P [(R_L + R_r + R_l)^2 + (X_L + X_A)^2]} \quad \dots (6.72)$$

6.14.3. Physical Aperture. Still another type of aperture is known as '**Physical Aperture**' which is related to the actual physical size (or cross-section) of the antenna and is denoted by the symbol A_p . This is why physical aperture is of more meaningful for the antennas of larger physical size or cross-section in term of wavelength like horn, parabolic reflector and mattress multi-element etc. i.e. for **some particular type of antennas only**, while effective aperture is a unique quantity for *any antenna*. Hence physical aperture may be defined as '**the physical cross-section perpendicular to the direction of propagation of incident electromagnetic wave with antenna set for maximum response**'.

In larger cross-section (in wavelength) antennas like horn, parabolic reflector, mattress multi-element, the physical aperture is greater than effective aperture which for antenna like short dipole, it is less than the effective aperture. For example, the value of effective aperture is approx. 50% of physical aperture, for horn antenna, between 50% to 65% for parabolic reflector and about cent per-cent for large mattress multi-element type of antenna. In an ideal size when there is no thermal losses and the field is in phase the physical aperture A_p and effective aperture A_e are equal and under this condition maximum directivity is achieved i.e.

$$A_p = A_e \dots \dots \text{when no losses} \quad \dots (6.73)$$

But directivity D and effective aperture, A_e are related by the

$$D = \frac{4\pi}{\lambda^2} \cdot A_e$$

$$D_{\max} = \frac{4\pi}{\lambda^2} \cdot A_p$$

From eqn. 6.73 ... (6.74)

Further, the ratio of maximum effective aperture to the physical aperture is given the name as absorption ratio and is denoted by γ . Thus

$$\text{Absorption ratio} = \gamma = \frac{(A_e)_{\max}}{A_p} \quad \dots (6.75)$$

The values of γ lies between 0 and ∞ and being ratio has no unit.

6.15. RELATION BETWEEN MAX. APERTURE AND GAIN OR DIRECTIVITY

(AMIEIE Dec. 1978, Nov. 1977)

The radiation pattern is same for transmitting and receiving antennas, by virtue of reciprocity theorem and hence the idea of directivity, which itself is related with shape of the radiation pattern, is extended for receiving antennas also. It is found, in practice, that the directivity of receiving antennas are directly proportional to the maximum effective apertures.

Let there be two antennas A and B whose directivities and maximum effective apertures are denoted by D_a , D_b and $(A_{ea})_{\max}$ and $(A_{eb})_{\max}$ respectively.

$$\begin{aligned} \therefore D_a &\propto (A_{ea})_{\max} \\ \text{and } D_b &\propto (A_{eb})_{\max} \\ \therefore \frac{D_a}{D_b} &= \frac{(A_{ea})_{\max}}{(A_{eb})_{\max}} \quad \dots (6.76) \end{aligned}$$

But from the eqn. 6.51, the gain and directivity w.r.t. isotropic source or antenna is given by

$$G_0 = kD$$

where

G_0 = Gain of a transmitting or receiving antenna.

k = Efficiency factor.

D = Directivity.

If now the losses of efficiency factor k and mismatch are included, the k can be replaced by effectiveness ratio α , i.e.

$$G_0 = \alpha D \quad \dots (6.77)$$

Let us now assume that G_{oa} , α_a , D_a being the gain, effectiveness ratio and directivity of antenna A and G_{ob} , α_b , D_b the corresponding quantities for the antenna B , then from eqn. 6.77.

$$G_{oa} = \alpha_a D_a \dots \dots \dots \text{for Ant. } A$$

and

$$G_{ob} = \alpha_b D_b \dots \dots \dots \text{" " } B$$

$$\therefore \frac{G_{oa}}{G_{ob}} = \frac{\alpha_a D_a}{\alpha_b D_b} = \frac{\alpha_a (A_{ea})_{\max}}{\alpha_b (A_{eb})_{\max}} \quad \text{from eqn. 6.76} \quad \dots (6.78)$$

But from definition,

$$\alpha_a = \frac{A_{ea}}{(A_{ea})_{\max}}$$

or

$$(A_{ea}) = \alpha_a (A_{ea})_{\max} \quad \dots 6.79 (a)$$

and similarly

$$(A_{eb}) = \alpha_b (A_{eb})_{\max} \quad \dots 6.79 (b)$$

Hence from eqn. 6.78

$$\frac{G_{oa}}{G_{ob}} = \frac{A_{ea}}{A_{eb}}$$

when A_{ea} and A_{eb} are the effectiveness apertures of antenna A and B . Let us now assume that antenna A is an isotropic antenna, then its directivity $D_a = 1$. Putting this condition in eqn. 6.76, we get

$$\frac{D_a}{D_b} = \frac{1}{D_b} = \frac{(A_{ea})_{\max}}{(A_{eb})_{\max}}$$

or

$$(A_{ea})_{\max} = \frac{(A_{eb})_{\max}}{D_b} \quad \dots 6.80 (a)$$

This eqn. suggests that if the maximum effective aperture and directivity of antenna (or any antenna) B are known, then the ratio of the two will give the maximum effective aperture of an isotropic antenna. Further, if the maximum effective aperture of an isotropic antenna is calculated, then the directivity of any antenna can be calculated from the following eqn. i.e.

$$D_b = \frac{(A_{eb})_{\max}}{(A_{ea})_{\max}} \quad \dots 6.80 (b)$$

This states that the directivity of any antenna is nothing but the *ratio of its max. effective aperture to the max. effective aperture of an isotropic antenna*.

For example, let us take the case of a short dipole antenna whose max. effective aperture and Directivity, if calculated, will be given by $\left(\frac{3}{8\pi}\right)\lambda^2$ and $\frac{3}{2}$, then from eqn. 6.80 (a).

$$\begin{aligned} (A_{ea})_{\max} &= \frac{3\lambda^2}{8\pi} = \frac{\lambda^2}{4\pi} \\ (A_{ea})_{\max} &= \frac{\lambda^2}{4\pi} \quad \dots (6.81) \end{aligned}$$

Putting this in eqn. 6.80 (b), the desired relation is obtained i.e.

$$D_b = \frac{(A_{eb})_{\max}}{\frac{\lambda^2}{4\pi}}$$

or

$$D_b = \frac{4\pi}{\lambda^2} (A_{eb})_{\max}$$

or in general

$$D = \frac{4\pi}{\lambda^2} (A_e)_{\max} \quad \dots 6.82 (a)$$

or simply

$$D = \frac{4\pi}{\lambda^2} A_{em} \quad \dots 6.82 (b)$$

This is the relation between directivity and max. effective aperture of an antenna.

6.16. EFFECTIVE LENGTH

(M.Sc. Phy. Lko. Univ. 1979, AMIETE Nov. 1977, Dec. 1981, 88, June 82, June 83)

The term 'effective length' of an antenna represents the effectiveness of an antenna as radiator or collector of electromagnetic wave energy. In other words, effective length indicates how far an antenna is effective in transmitting or receiving the electromagnetic wave energy.

For a receiving antenna, the effective length may be defined in terms of induced voltage v and incident field. *Effective length is nothing but the ratio of induced voltage at the terminal of the receiving antenna under open circuited condition to the incident Electric field intensity (or strength) E .* Thus.

$$\text{Effective length} = \frac{\text{Open circuited voltage}}{\text{Incident field strength (electric)}}$$

or

$$I_e = \frac{V}{E} \quad \text{metre or wavelength}$$

... (6.83)

l_e is in metre (m) or wavelength (λ) according to whether E is in volt/metre or volt/wavelength. Here V is the induced voltage under open circuited condition at the antenna terminals due to an uniform exciting field E .

Since induced voltage v also depends on the effective aperture and hence effective length and effective aperture of an antenna are related to each other as deduced below. From the eqn. 6.63 (a), the effective aperture is given by

$$A_e = \frac{V^2 R_L}{[(R_A + R_L)^2 + (X_A + X_L)^2] P}$$

or

$$V^2 = \frac{A_e \cdot [(R_A + R_L)^2 + (X_A + X_L)^2] P}{R_L} \quad \therefore P = \frac{E^2}{Z}$$

or

$$V = \frac{\sqrt{A_e [(R_A + R_L)^2 + (X_A + X_L)^2] E^2}}{\sqrt{Z R_L}}$$

$\therefore$

$$l_e = \frac{V}{E} = \frac{\sqrt{A_e [(R_A + R_L)^2 + (X_A + X_L)^2]}}{\sqrt{Z R_L}} \quad \dots 6.84 (a)$$

under conditions for maximum effective aperture, when

$$X_A = -X_L$$

$$R_A = R_r + R_l = R_L$$

or

$$R_A = R_r = R_L \text{ if } R_l = 0$$

$$l_e = \sqrt{\frac{(A_e)_m (2 R_r)^2}{\sqrt{Z R_r}}}$$

$$l_e = 2 \sqrt{\frac{(A_e)_m R_r}{\sqrt{Z}}} \quad \text{m or } \lambda$$

... 6.84 (b)

or

$$(A_e)_{\max} = \frac{l_e^2 Z}{4 R_r}$$

... 6.84 (c)

This is the relation between maximum effective aperture and effective length [eqn. 6.84 (b, c)].

Now for the transmitting antenna, the effective length is *that length of an equivalent linear antenna that has the same current $I(c)$ (as at the terminals of the actual antenna) at all the point along its length and that radiates the same field intensity E as the actual antenna.* This is illustrated in Fig. 6.8.

Here

$I(c)$ = Current at the terminals of actual antenna.

$I(z)$ = Current at any point Z of the antenna.

l_e = Effective length.

l = Actual length.

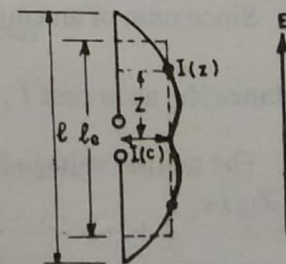


Fig. 6.8. Illustration of effective length for transmitting antenna.